



山东潍坊双星农药有限公司

SHANDONG WEIFANG SHUANGXING PESTICIDE CO., LTD.

MATERIAL SAFETY DATA SHEET

SECTION 1 Chemical product and Company identification

Product Name: 100g/L Chlorfenapyr SC

Product Use: Commercial insecticide.

Manufacturer: Shandong Weifang Shuangxing Pesticide Co., Ltd

(ADD: Binhai Development Zone, Weifang, Shandong, China)

SECTION 2 Composition/Information on Ingredients

Product Name	Wt.g/L	CAS RN
Chlorfenapyr	100	122453-73-0

SECTION 3 Hazards Identification

Emergency overview

Harmful if swallowed; Risk of serious damage to eyes. Potential health effects

Inhalation: Inhalation of product vapor may cause transient irritation of mucous membranes (nose, throat and respiratory tract), may cause headaches, dizziness, and drowsiness, could be anaesthetic, and may have other central nervous system effects lasting for up to 24 hours.

Ingestion: Symptoms include burning sensations and numbness in the mouth and throat, headache, dizziness, drowsiness, diarrhoea, nausea, vomiting, listlessness, stomach pain, muscular twitching of arms or legs, unconsciousness, convulsions, and coma (very high doses). Small amounts of liquid aspirated into the respiratory system during ingestion or from vomiting may cause bronchopneumonia or pulmonary oedema.

Skin contact: Irritating. Contact with the skin, especially the face, may result in initial stinging, burning or tingling sensations (fingertips, nose), followed by numbness or pain which may persist for up to 24 hours. (Although skin irritation tests in animals show cypermethrin to be non-irritating, experience has shown it to cause these symptoms on human skin.) Repeated exposure may cause skin dryness or cracking.

Eye contact: May be severely irritating to the eyes.



山东潍坊双星农药有限公司

SHANDONG WEIFANG SHUANGXING PESTICIDE CO., LTD.

Cancer information: N/A

Teratology: N/A

Reproductive effects: N/A

SECTION 4 First Aid Measures

Inhalation:	Inhalation of product vapor may cause transient irritation of mucous membranes (nose, throat and respiratory tract), may cause headaches, dizziness, and drowsiness, could be anaesthetic, and may have other central nervous system effects lasting for up to 24 hours.
Ingestion:	Symptoms include burning sensations and numbness in the mouth and throat, headache, dizziness, drowsiness, diarrhoea, nausea, vomiting, listlessness, stomach pain, muscular twitching of arms or legs, unconsciousness, convulsions, and coma (very high doses). Small amounts of liquid aspirated into the respiratory system during ingestion or from vomiting may cause bronchopneumonia or pulmonary oedema.
Skin contact:	Irritating. Contact with the skin, especially the face, may result in initial stinging, burning or tingling sensations (fingertips, nose), followed by numbness or pain which may persist for up to 24 hours. (Although skin irritation tests in animals show deltamethrin to be non-irritating, experience has shown it to cause these symptoms on human skin.) Repeated exposure may cause skin dryness or cracking. May be severely irritating to the eyes.
Eye contact:	May be severely irritating to the eyes.
Ingestion:	N/A
Notes to physician:	N/A
Antidote:	N/A

SECTION 5 Fire Fighting Measures

Hazards from combustion products

Heating/combustion will generate oxides of carbon and nitrogen, bromine compounds and other irritant and toxic fumes.

Precautions for fire fighters

The product is a Class C1 Combustible liquid. Firefighters should wear full protective gear, including self-contained breathing apparatus (AS/NZS 1715/1716). Keep unnecessary people away.



山东潍坊双星农药有限公司

SHANDONG WEIFANG SHUANGXING PESTICIDE CO., LTD.

If it can be done safely, remove intact containers from the fire. Otherwise, use water spray to cool them. Avoid spraying directly into containers due to danger of boil over. Bund area with sand or earth to prevent contamination of drains or waterways.

SECTION 6 Accidental release measures

Avoid contact with the spilled material or contaminated surfaces. Extinguish or remove all possible sources of ignition. When dealing with spills do not eat, drink or smoke and wear protective clothing and equipment as described in Section 8 - PERSONAL PROTECTION. Keep people and animals away. Prevent spilled material from entering drains or watercourses. Contain spill and absorb with earth, sand, clay, or other absorbent material. Collect and store in properly labeled, sealed drums for safe disposal. Deal with all spillages immediately. If contamination of drains, streams, watercourses, etc. is unavoidable, warn the local water authority.

SECTION 7 Handling and storage

Handling Keep out of reach of children. Will irritate the eyes and skin. Avoid contact with eyes and skin. Do not inhale vapor. If product in eyes washes it out immediately with water. After use and before eating, drinking or smoking, wash hands, arms and face thoroughly with soap and water. After each day's use, wash gloves, goggles and contaminated clothing.

Storage Store in the closed, original container in a cool, secure, well-ventilated area. Do not store for prolonged periods in direct sunlight. Keep away from all ignition sources.

Flammability: Combustible liquid, Class C1 - flashpoint between 61° C and 150° C.

SECTION 8 Exposure controls/Personal protection

Exposure standards Bayer recommends an Occupational Exposure Standard for cypermethrin of 0.28 g/m³. The manufacturer of the solvent recommends the following Occupational Exposure Limits:

Solvent naphtha (petroleum), heavy aromatic: TWA: 100 mg/m³ (17ppm).

For the naphthalene present in the solvent the NOHSC Occupational Exposure Limits are: TWA: 10ppm (52 mg/m³, STEL: 15ppm (79 mg/m³). Skin notation. Definitions:

Exposure standard – Time Weighted Average (TWA) means the average airborne concentration of a particular substance when calculated over a normal eight-hour working day, for a five-day working week.

Exposure standard – Short term exposure limit (STEL) means a 15 minute TWA exposure which should not be exceeded at any time during the working day.

Skin notation – Absorption through the skin may be a significant source of exposure. Engineering controls Control process conditions to avoid contact. Use in a well-ventilated area only.

Personal Protective Equipment



山东潍坊双星农药有限公司

SHANDONG WEIFANG SHUANGXING PESTICIDE CO., LTD.

It is important to protect eyes and skin as contact with this product may cause severe pain.

- _ Wear face shield
 - _ Wear cotton overalls buttoned to the neck and wrist and a washable hat.
 - _ Wear elbow-length PVC gloves.
 - _ If working in a poorly ventilated area or if occupational exposure levels are likely to be exceeded, wear a respirator suitable for organic vapors - AS/NZS 1715/1716 approved.
-

SECTION 9 Physical & Chemical properties

Appearance: white liquid

Odor: Distinctive aromatic hydrocarbon

pH: 5.0 to 8.0 (5% in water)

Vapor pressure: 4.6kPa at 38°C (hydrocarbon solvent)

Vapor density: > 1.0 (hydrocarbon solvent)

Boiling point: 443.5° C (hydrocarbon solvent)

Freezing/melting

Point: Not available

Solubility: Forms an emulsion in water

Specific Gravity: 1.53 at 20° C

Flash Point: 222° C (Pensky Martens Closed Cup)

Flammability (explosive) limits: LEL: 0.6; UEL: 7.0 Vol. % in air (hydrocarbon solvent)

Auto-ignition temperature: > 400° C

SECTION 10 Stability & Reactivity

Chemical stability: Stable under normal conditions of use.

Hazardous polymerization: Will not polymerize.

Conditions to avoid: Avoid sources of ignition and extreme heat.

Incompatible materials: Avoid oxidizing agents, strong acids and alkalis.

Hazardous decomposition products: Heating/combustion will generate oxides of carbon and nitrogen, and bromine compounds.

SECTION 11 Toxicological Information

Inhalation: Inhalation of product vapor may cause transient irritation of mucous membranes (nose, throat and respiratory tract), may cause headaches, dizziness, rowsiness, could be anaesthetic, and may have other central nervous system effects lasting for up to 24 hours.

Skin contact Irritating. Contact with the skin, especially the face, may result in initial stinging, burning or tingling sensations (fingertips, nose), followed by numbness or pain which may persist for



山东潍坊双星农药有限公司

SHANDONG WEIFANG SHUANGXING PESTICIDE CO., LTD.

up to 24 hours. (Although skin irritation tests in animals show cypermethrin to be non-irritating, experience has shown it to cause these symptoms on human skin.) Repeated exposure may cause skin dryness or cracking.

Eye contact May be severely irritating to the eyes.

Ingestion Symptoms include burning sensations and numbness in the mouth and throat, headache, dizziness, drowsiness, diarrhoea, nausea, vomiting, listlessness, stomach pain, muscular twitching of arms or legs, unconsciousness, convulsions, and coma (very high doses). Small amounts of liquid aspirated into the respiratory system during ingestion or from vomiting may cause bronchopneumonia or pulmonary oedema.

ANIMAL TOXICITY DATA – PRODUCT

Acute:

Oral toxicity LD₅₀ rat: 6480 mg/kg (similar product)

Dermal toxicity LD₅₀ rabbit: > 2000 mg/kg (similar product)

Inhalation toxicity LC₅₀ rat: > 11.3 mg/L (6 h) (similar product)

Skin irritation Slightly irritating (rabbit) (similar product)

Eye irritation Irritating (rabbit) (similar product)

Sensitisation Non-sensitizing (guinea pig) (similar product)

Chronic: Chlorfenapyr is not mutagenic, carcinogenic or teratogenic. A National Toxicology Program (NTP) report states that lifetime inhalation exposure to naphthalene resulted in increases in tumors of the nose in rats. In a previous NTP study, lifetime inhalation exposure to naphthalene increased lung tumors in female mice. The relevance of the rodent findings to humans is questionable.

SECTION 12 Ecological Information

Chlorfenapyr is highly toxic to fish, aquatic organisms and bees in laboratory studies, but considered less toxic under natural conditions. DO NOT contaminate streams, rivers or waterways with Cypermethrin Options or the used containers.

Ecotoxicity Chlorfenapyr:

Fish toxicity: LC₅₀ (96 h) 0.001 – 0.01 mg/L

Bird toxicity: Acute oral LD₅₀ for duck > 5000 mg/kg Acute oral

LD₅₀ for quail > 10000 mg/kg

Environmental fate, persistence and degradability

Chlorfenapyr is not persistent in the environment. It is destroyed by soil micro-organisms and does not leave residues in the environment or build up in the food chain. The hydrocarbon solvent is volatile and will evaporate if released into water. It is biodegradable.



山东潍坊双星农药有限公司

SHANDONG WEIFANG SHUANGXING PESTICIDE CO., LTD.

SECTION 13 Disposal Considerations

When returnable container is empty or contents no longer required return it to the point of purchase. For non-returnable containers, triple or (preferably) pressure rinses them before disposal. Add rinsing to spray tank. Do not dispose of undiluted chemicals on-site. If recycling, replace cap and return clean containers to recycler or designated collection point. If not recycling, break, crush or puncture and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt. Dispose of waste product through a reputable waste contractor.

SECTION 14 Transport Information

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

Not hazardous under the applicable DOT, ICAO/IATA, IMO, TDG and Mexican regulations.

UN No	3082
Proper shipping name	Pesticide, Liquid, Toxic, N.O.S. (Chlorfenapyr)
Packing group	III
Class	9

SECTION 15 Regulatory Information

Registered according to the Agricultural and Veterinary Chemicals Act 2005 National Registration Authority approval number: 51849

SECTION 16 Other Information

This MSDS summarizes our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this MSDS and consider the information in the context of how the product will be handled and used in the workplace including in conjunction with other products. If clarification or further information is needed to ensure that an appropriate risk assessment can be made, the user should contact this company. Our responsibility for products sold is subject to our standard terms and conditions, a copy of which is sent to our customers and is also available on request.